IPL Cricket Analysis & Match Prediction System

This project develops a complete IPL Cricket Analysis and Prediction system using data science techniques.

The system performs data preprocessing, exploratory data analysis, feature engineering, and predictive modeling

to uncover team and player performance patterns and forecast match outcomes.

Dataset:

The project uses two datasets: matches.csv and deliveries.csv. These datasets include match-level and

ball-by-ball information such as teams, venues, toss results, runs scored, wickets, and player performance.

Data Preprocessing:

Missing values were handled using appropriate imputation techniques. Inconsistent team names were standardized,

and new features such as total runs, strike rates, and economy rates were created to enhance predictive performance.

Exploratory Data Analysis:

EDA was performed to identify trends including winner prediction by year, super over frequency, umpire trends,

high-scoring matches (200+ runs), player comparisons, and season-wise Orange and Purple Cap winners.

Visualizations such as bar charts, heatmaps, and line plots were used to present insights.

Predictive Modeling:

Multiple classification models including Logistic Regression, Decision Tree, and Random Forest were implemented.

Random Forest provided the best accuracy for predicting match outcomes using features such as teams, toss winner,

and venue.

Results:

The analysis revealed key insights such as the influence of toss outcomes on match results, increasing scoring trends,

and consistent dominance of certain players across seasons. The predictive model achieved strong accuracy and provides

valuable decision-support for analysts and fantasy league players.

Conclusion:

This project demonstrates how sports analytics combined with machine learning techniques can generate actionable

insights and predictive intelligence for cricket performance analysis.